

Name: SOLUTIONS

Circle your teacher: CHESHIRE FRIDAY HILL LE LOH STAFFE

Materials allowed: Formulae sheet

Show full working to obtain full marks for each of the questions in this section.

Time: 20 minutes

Marks: / 30

Question 1 (4 marks: 2, 2)

Solve for the unknown in the following linear equations.

a) $17 = 3(3x - 2) - 4$

$$17 = 9x - 6 - 4$$

$$17 = 9x - 10$$

$$27 = 9x$$

$$\underline{\underline{3 = x}}$$

✓ working (expands bracket)
✓ solve

b) $\frac{-b}{3} - 10 = 3b$

$$-b - 30 = 9b$$

$$-30 = 10b$$

$$\underline{\underline{-3 = b}}$$

✓ removes denominator
✓ solve

Question 2 (6 marks: 3, 3)

Solve the following simultaneous equations algebraically:

a) $y = 5x - 2 \dots (1)$
 $7x - y = 2 \dots (2)$

sub (1) into (2)

$$7x - (5x - 2) = 2$$

$$7x - 5x + 2 = 2$$

$$2x = 0$$

$$\underline{\underline{x = 0}}$$

✓ sub or eliminate

✓ one variable

sub $x = 0$ into (1)

$$y = 5(0) - 2$$

$$\underline{\underline{y = -2}}$$

$$\therefore \underline{\underline{x = 0, y = -2}}$$

✓ Both variables

b) $4x + 3y = 11 \dots (1)$
 $3x + 2y = 8 \dots (2)$

$$(1) \times 3$$

$$12x + 9y = 33 \dots (3)$$

$$(2) \times 4$$

$$12x + 8y = 32 \dots (4) \text{ (or rearrange)}$$

✓ scale eqns

$$(3) - (4)$$

$$\underline{\underline{y = 1}}$$

✓ solves for one variable

sub $y = 1$ into (1)

$$4x + 3(1) = 11$$

$$4x = 8$$

$$\underline{\underline{x = 2}}$$

$$\therefore \underline{\underline{x = 2, y = 1}}$$

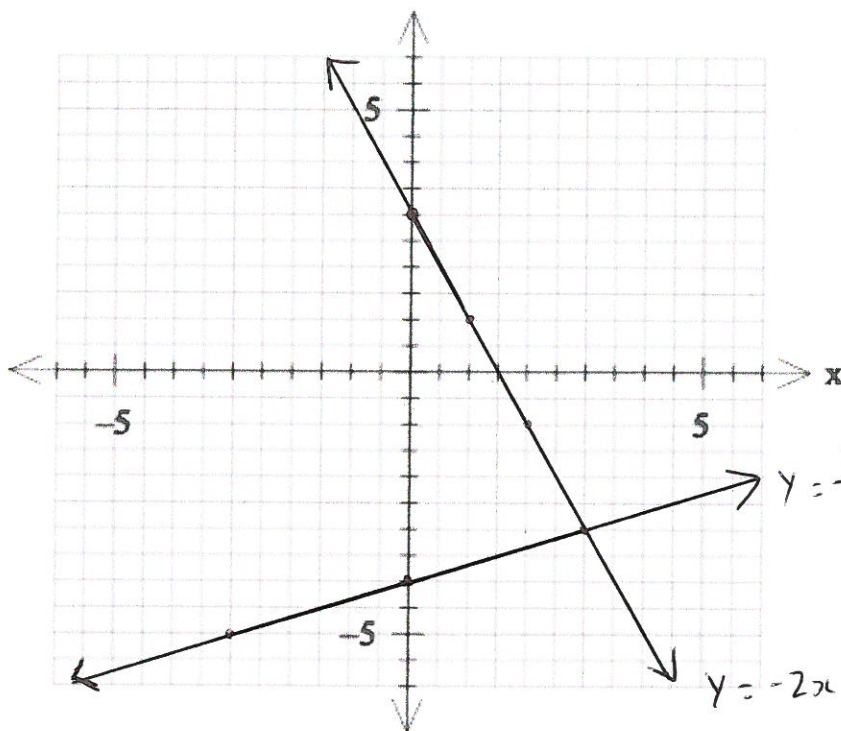
✓ solves for both variables / 10

Question 3 (5 marks: 2, 2, 1)

a) Graph both of the following equations on the Cartesian plane below.

i. $y = \frac{x}{3} - 4$

ii. $y - 3 + 2x = 0$ $y = -2x + 3$



✓ y-int
✓ draw accurate with ruler and arrows included

✓ y-int
✓ accurate line with ruler and arrows

b) Use your graph to solve the simultaneous equations $y = \frac{x}{3} - 4$ and $y - 3 + 2x = 0$

$x = 3, y = -3$ or $(3, -3)$ ✓ correct point from graph

Question 4 (4 marks)

Adam is currently 20 years younger than Brian. In two years time, Brian will be twice as old as Adam. Using one variable, determine the current age of Adam and Brian.

	Brian	Adam
Now	x	$x - 20$
In 2 years	$x + 2$	$x - 18$

Let "x" represent Brian's current age
✓ define variable

$$x + 2 = 2(x - 18)$$

✓ equation w/ one variable

$$x + 2 = 2x - 36$$

$$x = 38$$

✓ solve

Adam is currently 18 and
Brian is currently 38. 2

✓ statement

Question 5 (11 marks: 2, 3, 4, 1, 1)

The table below shows the battery charge remaining in a mobile phone x hours after being unplugged from the charger.

Time (x hours)	0	1	2	3
Charge (y %)	88	82	76	70

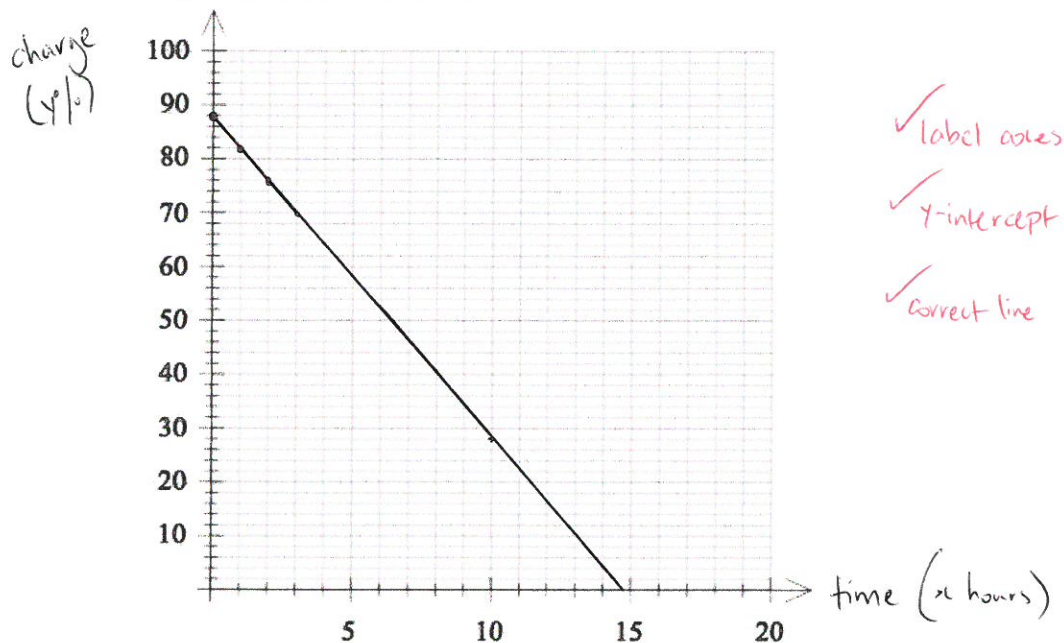
- a) Using the information above, write an equation to represent the battery charge remaining (y %) in a mobile phone x hours after being unplugged from the charger.

$$y = -6x + 88$$

✓ correct gradient

✓ correct y-intercept

- b) Graph the equation below. Label all axes.



✓ label axes

✓ y-intercept

✓ correct line

- c) Find and interpret the y-intercept and gradient of the line.

y-intercept: $(0, 88)$ ✓ → initial phone charge ✓

Gradient: -6 ✓ → phone loses 6% charge per hour ✓

- d) Use your equation to determine the battery charge remaining after 4.5 hours?

$$y = -6(4.5) + 88$$

Remaining charge will be 61% after 4.5 hours

$$= 61\%$$

✓ solve uses equation

- e) Use your graph to estimate how long, to the nearest hour, it will take for the phone to run out of charge?

It will take the phone 15 hours to run out of charge.

✓ estimates 15 hours using graph

Name: _____

Circle your teacher: CHESHIRE FRIDAY HILL LE LOH STAFFE

Materials allowed: Formulae sheet, calculator, Classpad, 1 x A4 page notes
Show full working to obtain full marks for each of the questions in this section.

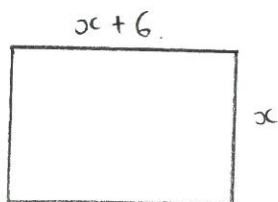
Time: 30 minutes

Marks: / 33

Question 6 (8 marks: 4, 4)

Write an equation to clearly represent the following situations before solving it.
Clearly define your variables.

a) Sabrina is a landscape gardener who is currently working on a rectangular piece of garden. The length of the garden is 6 metres longer than the width, and the perimeter of the garden is 62 metres. Calculate the area of the garden.



Let 'x' be the width of the garden.

$$2x + 2(x+6) = 62$$

✓define variable

$$2x + 2x + 12 = 62$$

✓write equation

$$4x = 50$$

$$x = 12.5$$

✓solve for width

$$\begin{aligned} \text{Area} &= 12.5 \times 18.5 \\ &= 231.25 \end{aligned}$$

∴ The area of the garden is 231.25 m^2 . ✓writes statement

b) I think of a number, double it and add eleven. I find that if I divide this result by three, I end up with one less than the number I first thought of. Determine the number I first thought of.

Let 'x' be the number I first thought of.

$$\frac{2x+11}{3} = x-1$$

✓define variable

✓write equation

$$2x+11 = 3(x-1)$$

$$2x+11 = 3x-3$$

$$x = 14$$

✓solve for number

∴ The number I first thought of was 14.

✓writes statement

Question 7 (9 marks: 1, 2, 2, 2, 2)

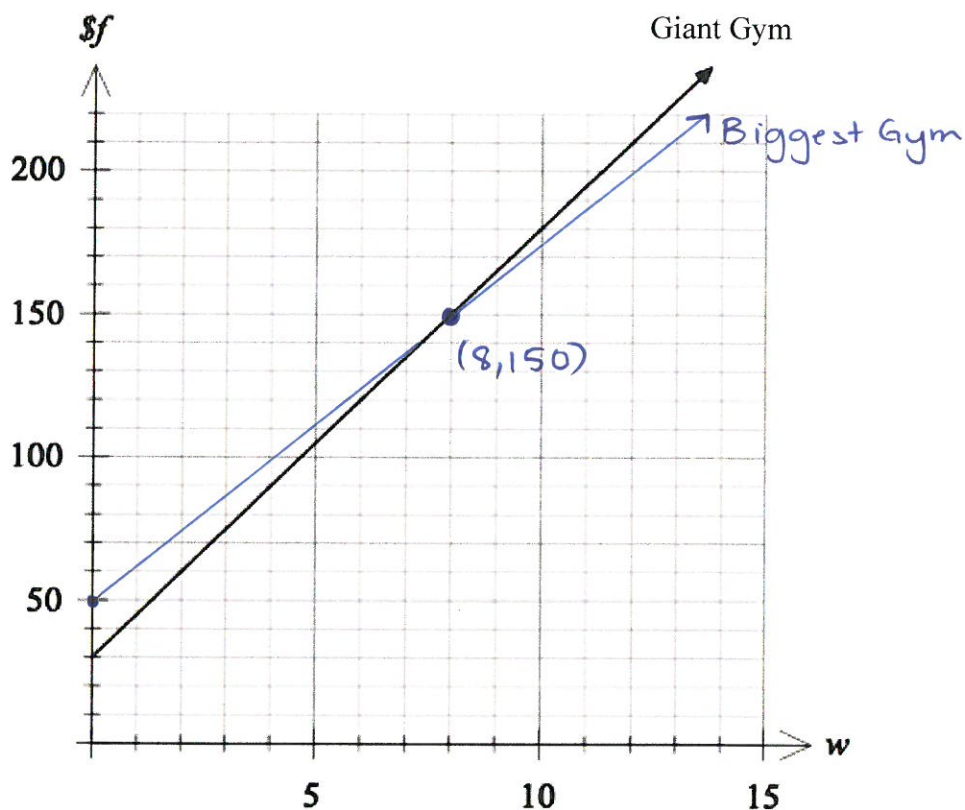
John has just finished school and he is thinking about joining a gym.

He does some research and finds that Biggest Gym charges a joining fee of \$50 and a weekly fee of \$12.50.

- a) Write an equation for the total fees charged f (\$) in terms of the number of weeks of membership (w) for Biggest Gym.

$$f = 12.50w + 50 \quad \checkmark \text{ writes correct equation}$$

The second Gym that John considers joining is Giant Gym. The total fees charged f (\$) in terms of the number of weeks of membership (w) for Giant Gym can be seen below.



- b) What is the joining fee and weekly fee charged by Giant Gym?

Giant Gym charges a joining fee of \$30

and a weekly fee of \$15. $\checkmark$ correct joining fee
 $\checkmark$ correct weekly fee

- c) On the axes above, plot the total fees charged f (\$) in terms of the number of weeks of membership (w) for Biggest Gym. Clearly label your line and identify key points.

$\checkmark$ correct line with arrow

$\checkmark$ uses a ruler and labels point of intersection & "Biggest Gym"

- d) State the coordinates of the point of intersection of the two lines from the graph and explain the significance of the intersection point in this context.

$(8, 150)$ ✓ states point of intersection
For each gym, 8 weeks of membership costs a total of \$150.

✓ explains context of point of intersection
* keyword: "each gym"

- e) John is unsure about the length of time that he would like to join the gym. By considering cost over time, provide a recommendation for which gym John should join.

✓ states Giant Gym for < 8 weeks
If John is considering joining the gym for less than 8 weeks, it is cheaper to join Giant Gym.

✓ states Biggest Gym for > 8 weeks
If he is considering joining for greater than 8 weeks, it works out cheaper to join Biggest Gym.

Question 8 (5 marks: 3, 2)

- a) Determine the equation of a straight line which passes through the points $(6, -1)$ and $(-3, 20)$.

$$m = \frac{20 - (-1)}{-3 - 6}$$

$$= -\frac{21}{9}$$

$$= -\frac{7}{3} \quad \text{✓ calculates gradient} \quad \therefore y = -\frac{7}{3}x + 13$$

$$y = -\frac{7}{3}x + c$$

sub $(6, -1)$

$$-1 = -\frac{7}{3}(6) + c$$

$$c = 13 \quad \text{✓ calculates y-int}$$

✓ state eqn of line

- b) Show working to determine whether or not the point $(18, 28)$ lies on the straight line above.

sub $(18, 28)$

$$28 = -\frac{7}{3}(18) + 13 \quad \text{✓ substitutes}$$

$$28 \neq -29$$

$$\therefore (18, 28) \text{ doesn't lie on the line } y = -\frac{7}{3}x + 13.$$

✓ concluding statement

Question 9 (4 marks)

Coastal Ice-creams sells ice-cream in single cones and double cones. The price of a single cone is \$4.30 whereas the price of a double cone is \$5.80. On a particular day, they sold a total of 400 cones and their earnings for that day was \$1990. How many of each type of cone did Coastal Ice-creams sell on this particular day?

Let 'x' be the number of single cones sold.

Let 'y' be the number of double cones sold.

$$4.30x + 5.80y = 1990 \dots (1)$$

✓define variables

$$x + y = 400 \dots (2)$$

✓write equations

$$x = 220 \quad y = 180 \quad \text{✓solve}$$

∴ Coastal Ice-creams sold 220 single cones

and 180 double cones on that day.

✓writes statement

Question 10 (7 marks: 4, 2, 1)

Nicole has a day off to run some errands. At 8am, she starts the day by driving at an average speed of 45km/h for 10 minutes to get to her closest mechanic where she leaves her car to get serviced. She walks home and arrives at her house 1 hour later. She then rides her bicycle and gets to the supermarket that is 2km away in 5 minutes. She spends 30 minutes shopping then rides her bicycle home at the same pace.

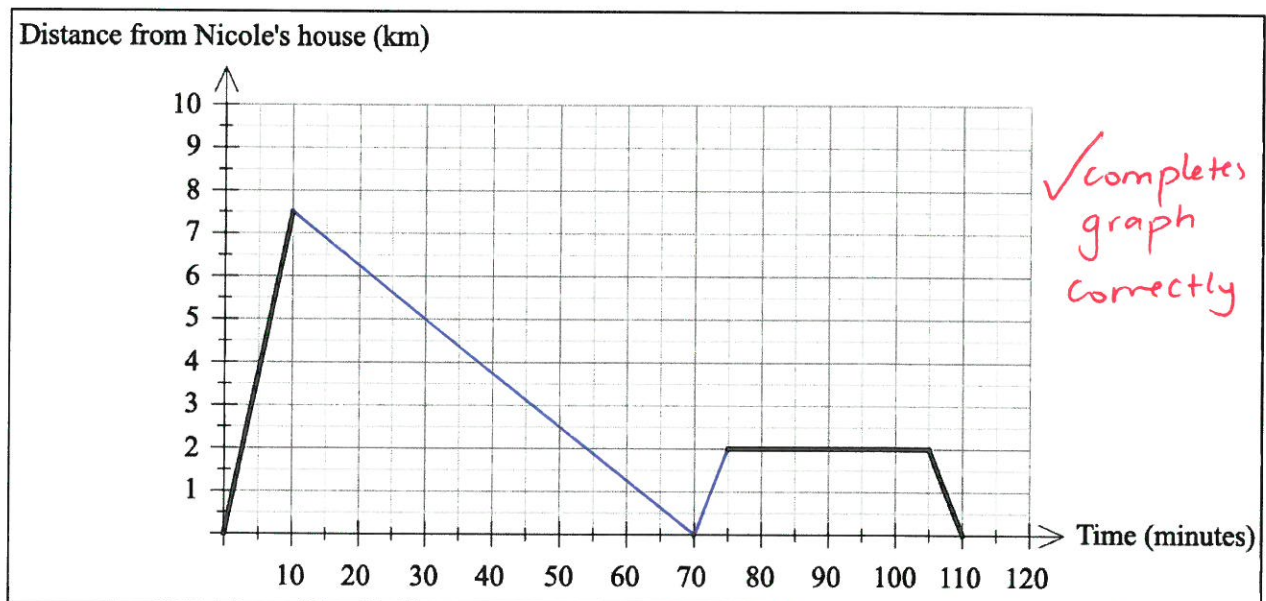
- a) Complete the intervals and rules for the piece-wise defined function and complete the graph for Nicole's day of errands:

Where time is t minutes after 8am and distance is d kilometres from Nicole's house,

✓ 2 eqns correct
✓ all correct

$$d = \begin{cases} \frac{0.75t}{-0.125t + 8.75} & 0 < t \leq 10 \\ \frac{0.4t - 28}{2} & 10 < t \leq 70 \\ \frac{-0.4t + 44}{-0.4t + 44} & 70 < t \leq 75 \\ \frac{75}{105} < t \leq \frac{105}{110} & 75 < t \leq 105 \end{cases}$$

✓ correct domain



- b) Use the equation(s) above to determine the time(s) at which Nicole was exactly 5km from her house to the nearest minute.

$$5 = 0.75t$$

$$5 = -0.125t + 8.75$$

✓ shows use of an equation

$$t \approx 7 \text{ mins}$$

$$t = 30 \text{ mins}$$

∴ Nicole was exactly 5km from home at 8:07am and 8:30am. ✓ both correct times

- c) Using the graph above, state the time interval, with justification, for which Nicole's average speed was the greatest.

Nicole's average speed was the greatest between 0 and 10 mins - this interval has the steepest gradient. ✓ states correct interval with justification